

INTRODUCTION TO INFO. & COMM. TECHNOLOGY

15-11-2023

Downloading Dev C++

- Contact Mr. Osama Tariq in SCIT for support regarding the installation of Dev C++ in your systems so you will be able to practice at home.

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Starting a new Source File

File

New

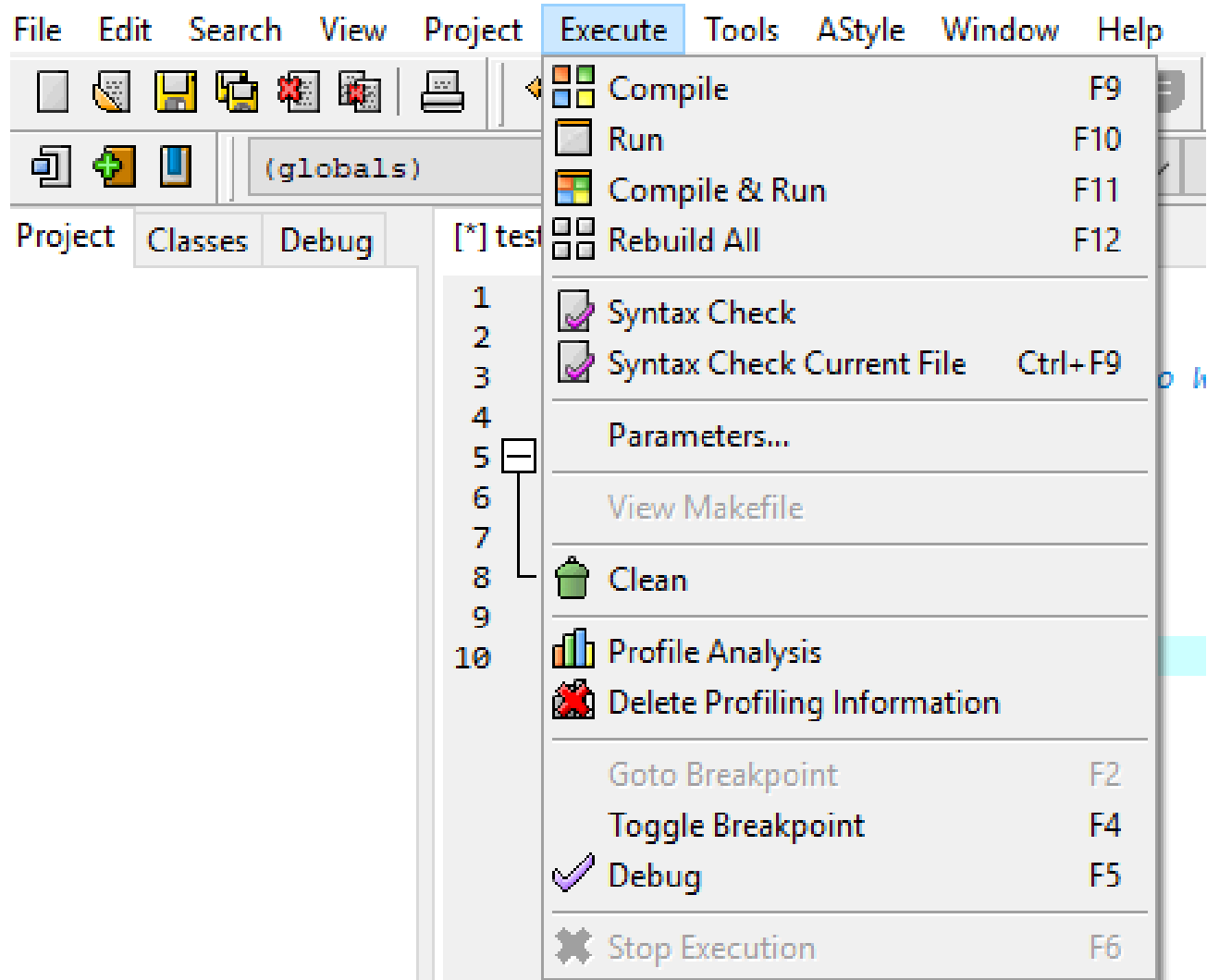
Source File

Your first file

[*] test.cpp

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout<<"Hello World";
6      return 0;
7  }
8
9  |
```

Compile & Run



#include <iostream>

- The word include with '#' tells the compiler to include the file `iostream` into the program
- File `iostream` is a header file needed for input/ output requirements of the program
- It is called a `preprocessor directive`
- The preprocessor reads your program before it is compiled and only executes those lines beginning with a `#` symbol
- The preprocessor is a program that “sets up” your source code for the compiler

using namespace std;

- All the elements of the standard C++ library are declared within std
- Programs usually contain several items with unique names, variables
- C++ uses *namespaces* to organize the names of program entities
- The statement `using namespace std;` declares that the program will be accessing entities whose names are part of the namespace called `std`

int main ()

- When program is loaded in the memory, the control is handed over to function main () and it is the first function to be executed

return 0;

- When a program finishes running, it sends a value to the operating system
- This returns the value of 0 to the operating system, which means that the program ended successfully

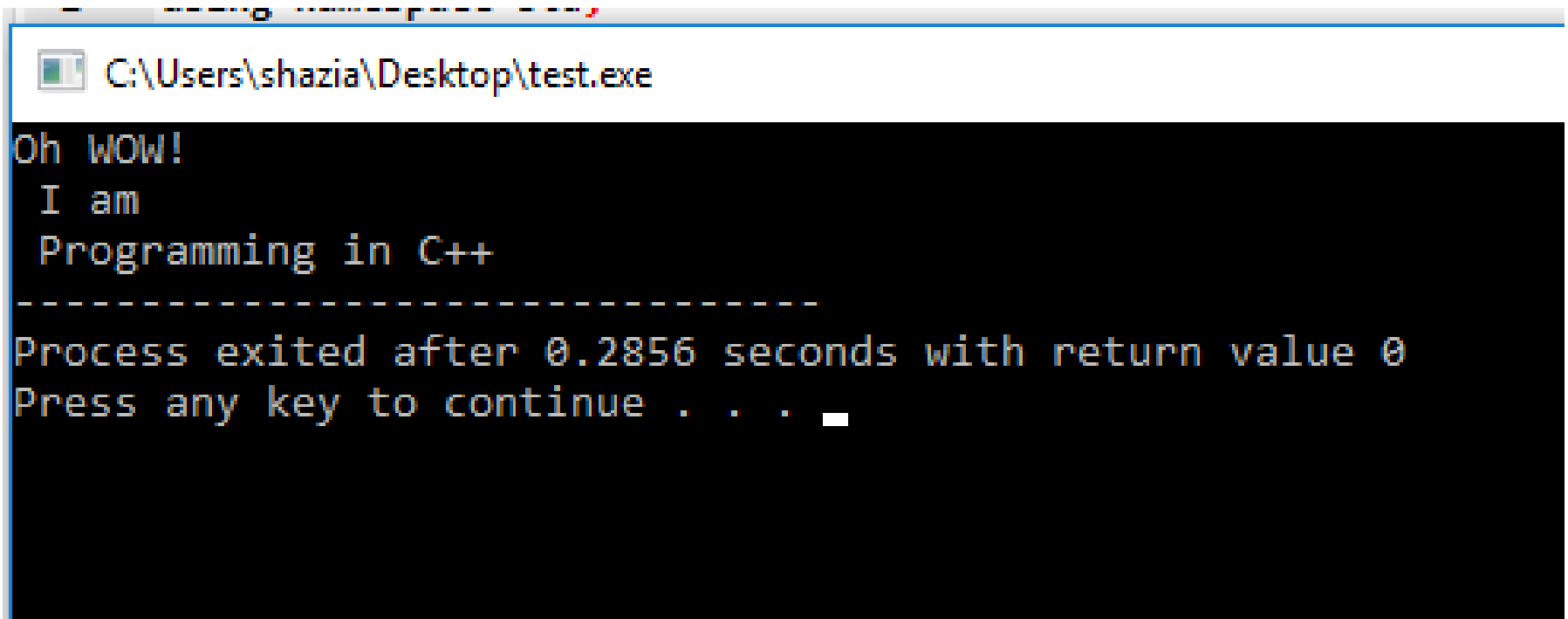
Statement terminator

- Semicolon is the statement terminator in C++

[*] test.cpp

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout<<"Hello World";
6      return 0;
7  }
8
9  |
```

Ex-1:Change the program to display:



```
C:\Users\shazia\Desktop\test.exe
Oh WOW!
I am
Programming in C++
-----
Process exited after 0.2856 seconds with return value 0
Press any key to continue . . .
```

\n is the new line character
Try to achieve the above

Answer:

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      cout << "Oh WOW! \n I am \n Programming in C++";
6      return 0;
7  }
8
```

Ex-2:

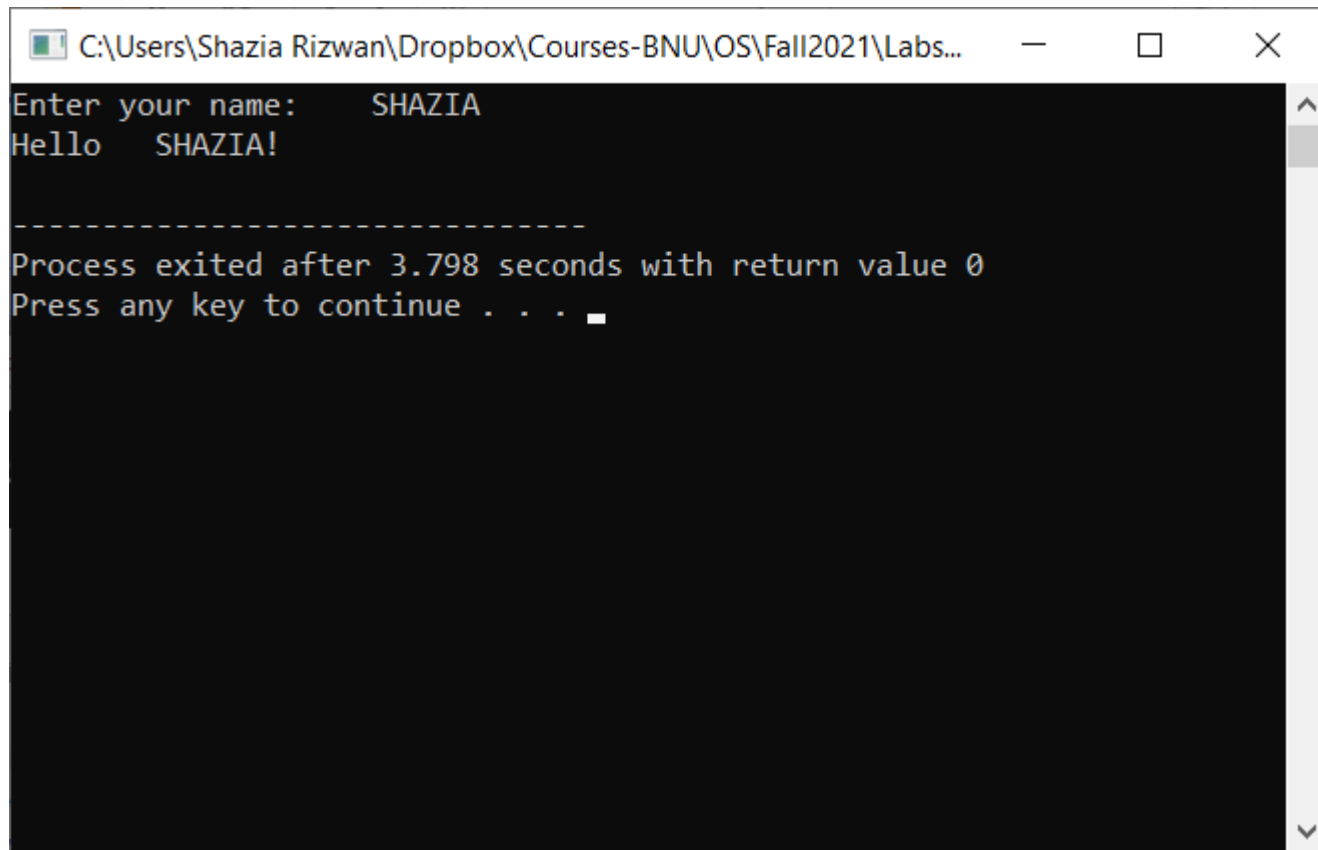
- Write a program which says Hello to you. Use data type `string`

Answer:

P1-20Dec.cpp

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      string name;
6      cout << "Enter your name: ";
7      cin>> name;
8      cout << "Hello " << name << "!\n";
9      return 0;
10 }
```

Output window



A screenshot of a Windows command prompt window. The title bar shows the file path: C:\Users\Shazia Rizwan\Dropbox\Courses-BNU\OS\Fall2021\Labs... The window has standard minimize, maximize, and close buttons. The command prompt shows the following text: "Enter your name: SHAZIA", "Hello SHAZIA!", a separator line of dashes, "Process exited after 3.798 seconds with return value 0", and "Press any key to continue . . .". A cursor is visible at the end of the last line.

```
C:\Users\Shazia Rizwan\Dropbox\Courses-BNU\OS\Fall2021\Labs...  
Enter your name:  SHAZIA  
Hello  SHAZIA!  
-----  
Process exited after 3.798 seconds with return value 0  
Press any key to continue . . .
```

Example-1

Find the
output ?

```
#include <iostream>
using namespace std;
int main()
{
    int num1, num2;
    double sale;
    char first;
    string str;
    num1 = 4;
    cout << "num1 = " << num1 << endl;
    num2 = 4 * 5 - 11;
    cout << "num2 = " << num2 << endl;
    sale = 0.02 * 1000;
    cout << "sale = " << sale << endl;
    first = 'D';
    cout << "first = " << first << endl;
    str = "It is a sunny day.";
    cout << "str = " << str << endl;
    return 0;
}
```


Output - Example-1

```
num1 = 4
num2 = 9
sale = 20
first = D
str = It is a sunny day.

-----
Process exited after 0.07283 seconds with return value 0
Press any key to continue . . .
```

Ref Chapters

- Book 3
- Chp 1,2